GHRaisoni International Skill Tech University Pune

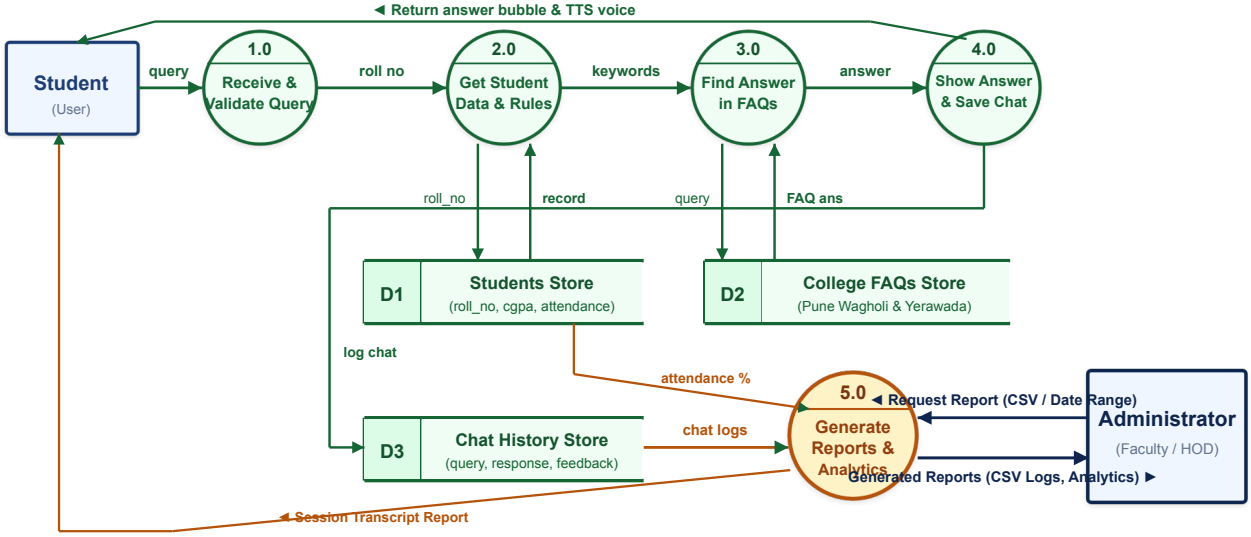
AI CHATBOT FOR COLLEGE

FACULTY REVISED SYSTEM ARCHITECTURE & DFD SPECIFICATION

✓ Context-Level DFD (Level 0) Included ✓ Standard Open-Ended Data Stores ✓ Circle Process Bubbles (DeMarco/Gane-Sarson)

Data Flow Diagram (Level 1: First-Level Process Decomposition & 5.0 Report Generation)

First-Level Data Flow Diagram (DFD Level 1: Process Decomposition)



DFD Symbols and Their Uses (Standard Gane-Sarson & DeMarco Notations)

Symbol	Name	Standard Academic Definition	Implementation in Our Project
Student	**External Entity**	A person, department, or system outside the software boundary that feeds or receives data (Square/Rectangle).	**Student (User) & Administrator** Students in Pune campuses, requests attendance, generates reports.
5.0 Reports	**Process (Bubble)**	An operation that transforms input data into output data. Standard notation is a circle with a process number and label.	**5.0 Generate Reports & Analytics** Compiles query logs CSV, attendance detention summaries.
D1 Store	**Data Store** *(Open-Ended)*	A repository where data is held persistently at rest. **MANDATORY RULE: Must use open-ended parallel lines (NEVER a closed box).**	**D1 Students, D2 College FAQs, D3 Chat History** Database tables supporting query matching and reports.
→	**Data Flow**	Directed solid arrow indicating the path of data in motion between entities, processes, and stores.	**query, record, FAQ answer, generated reports** Directional arrows carrying specific payload packages.

Fig: First-Level Data Flow Diagram (DFD Level 1) — Process decomposition into 1.0 to 4.0 Query Handlers & 5.0 Generate Reports & Analytics, standard open-ended parallel lines for data stores (D1 Students Store, D2 College FAQs Store, D3 Chat History Store), and full Gane-Sarson / DeMarco symbols table.